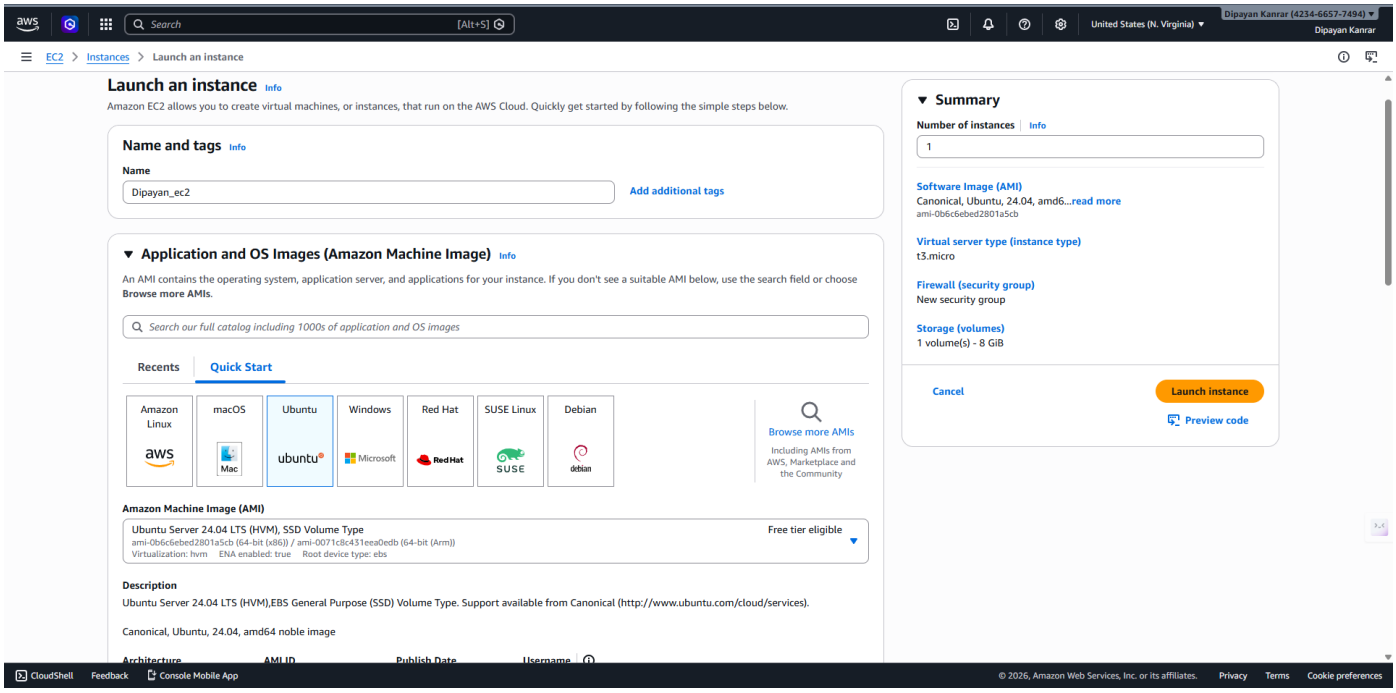
Assignment-9

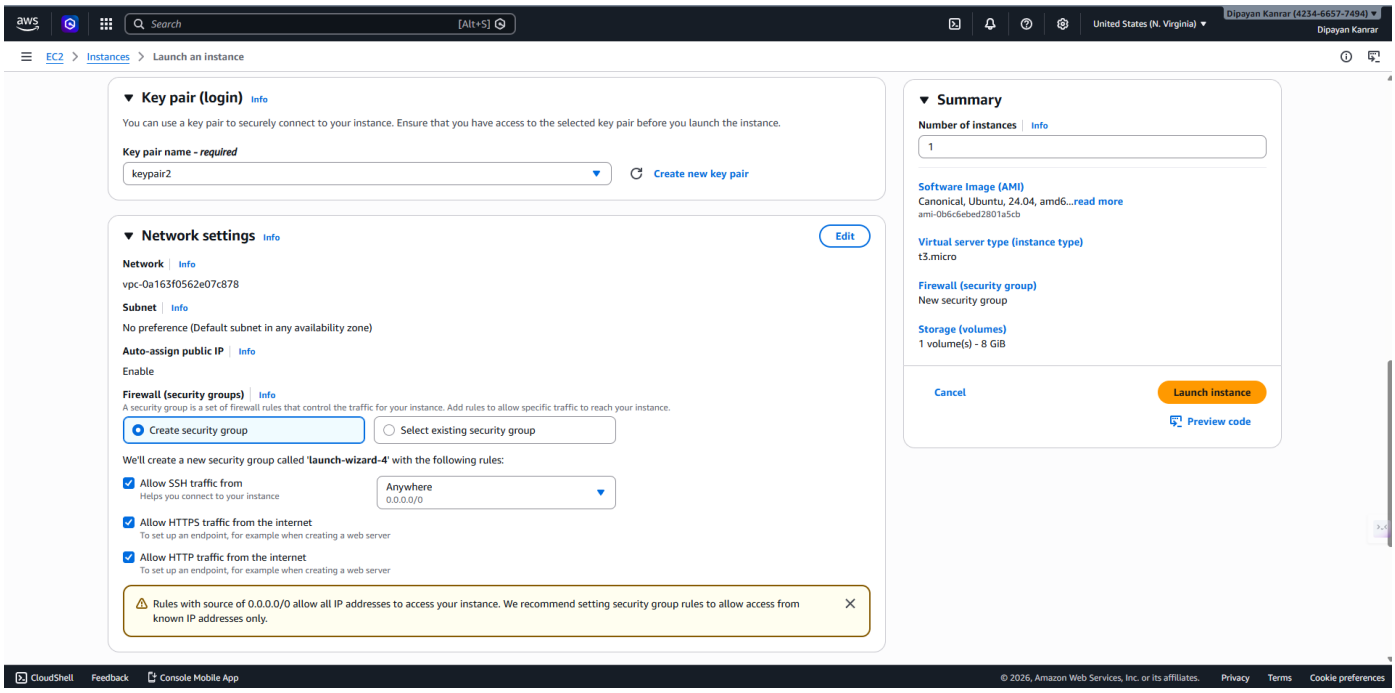
PROBLEM STATEMENT: Deploy a project from GitHub to EC2.

Steps to deploy project:

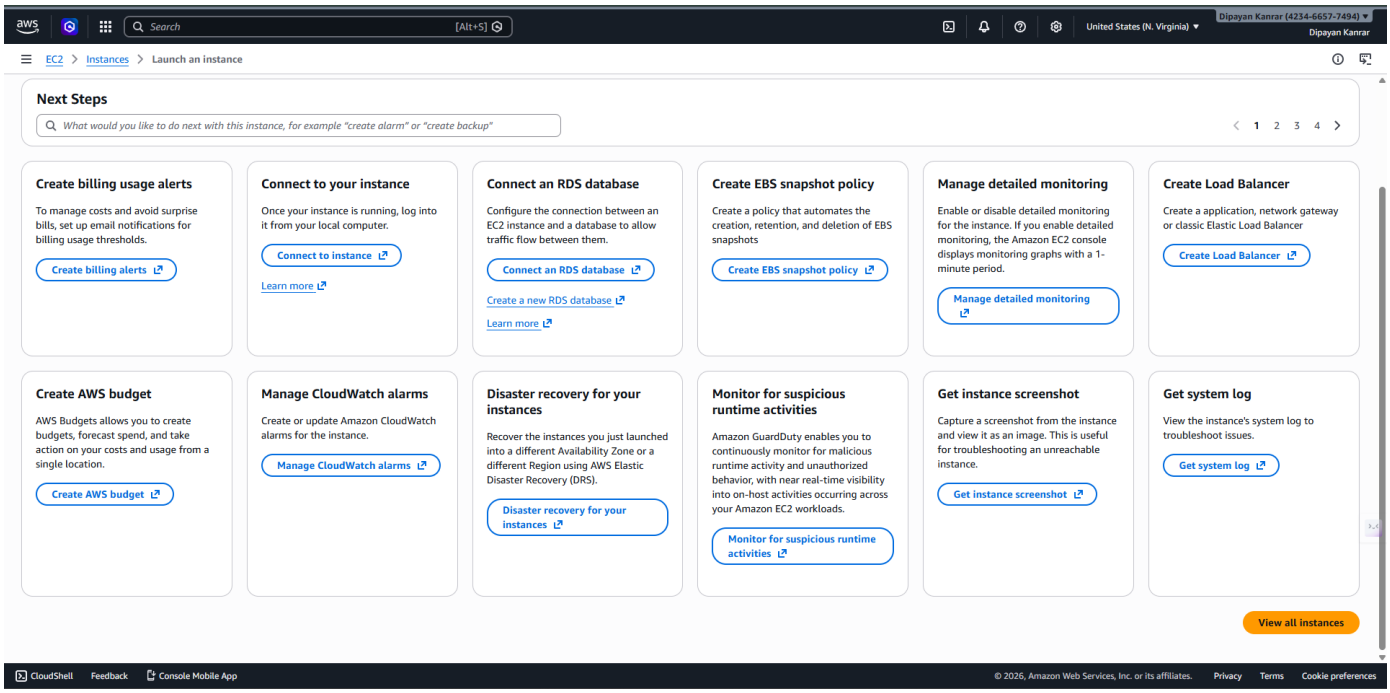
1. Sign up for an AWS account, search for 'EC2', select it and click on launch instance with Fill up the required Details →'Name', in 'Application and OS Images' select 'ubuntu'.



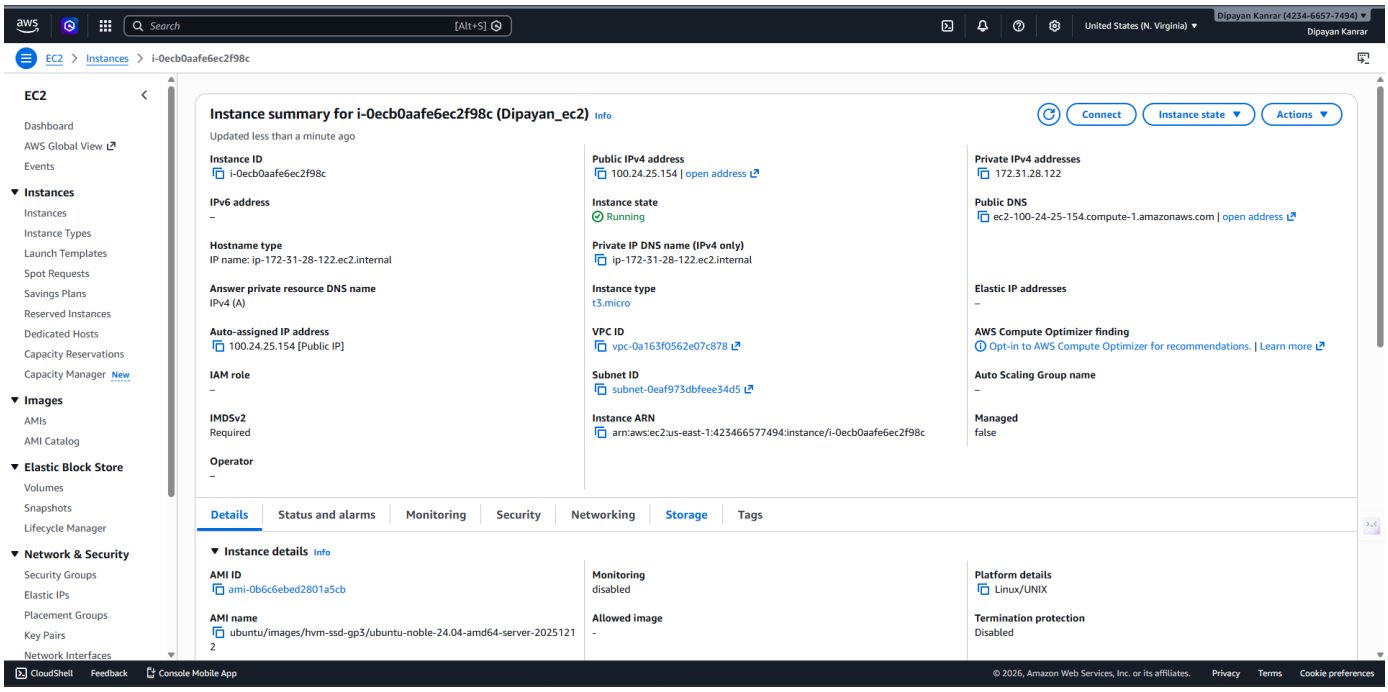
2. Under 'Network settings', tick off all the three checkboxes then click on 'Launch instance'.



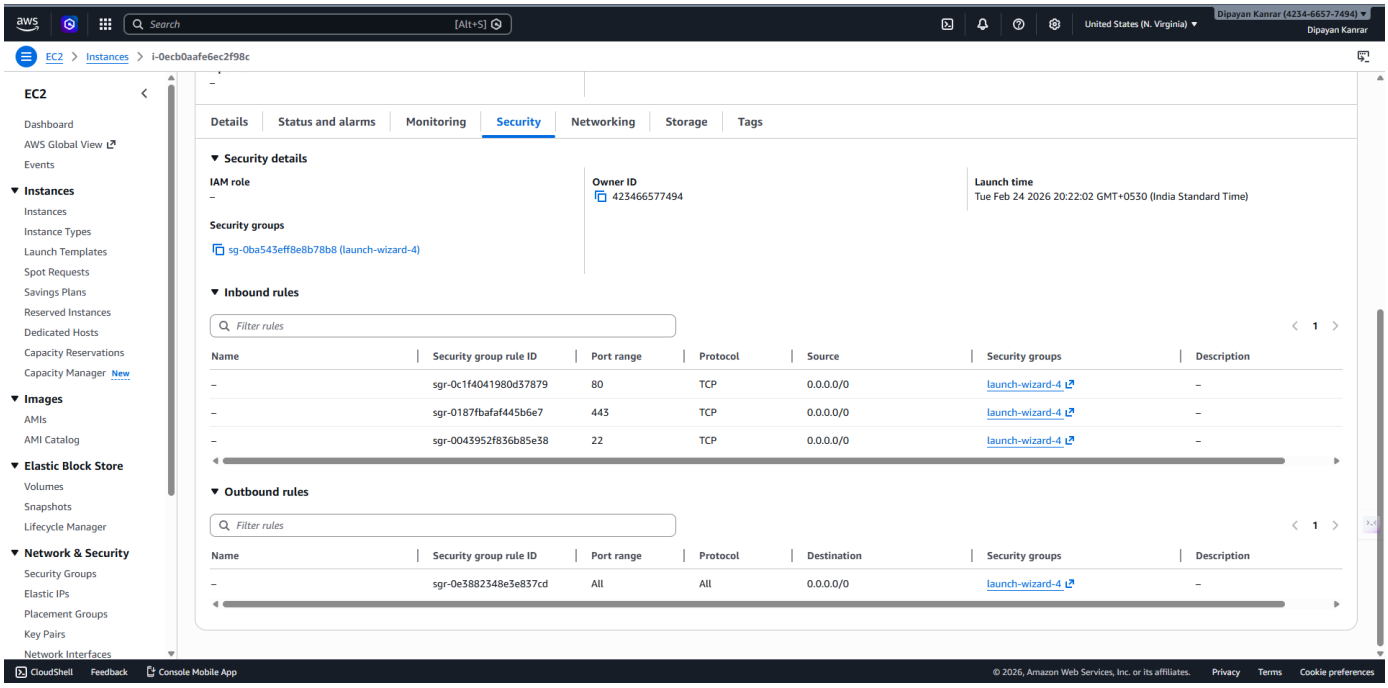
3.” Dipayan_ec2” instance is successfully created and then click on ‘Instances’.



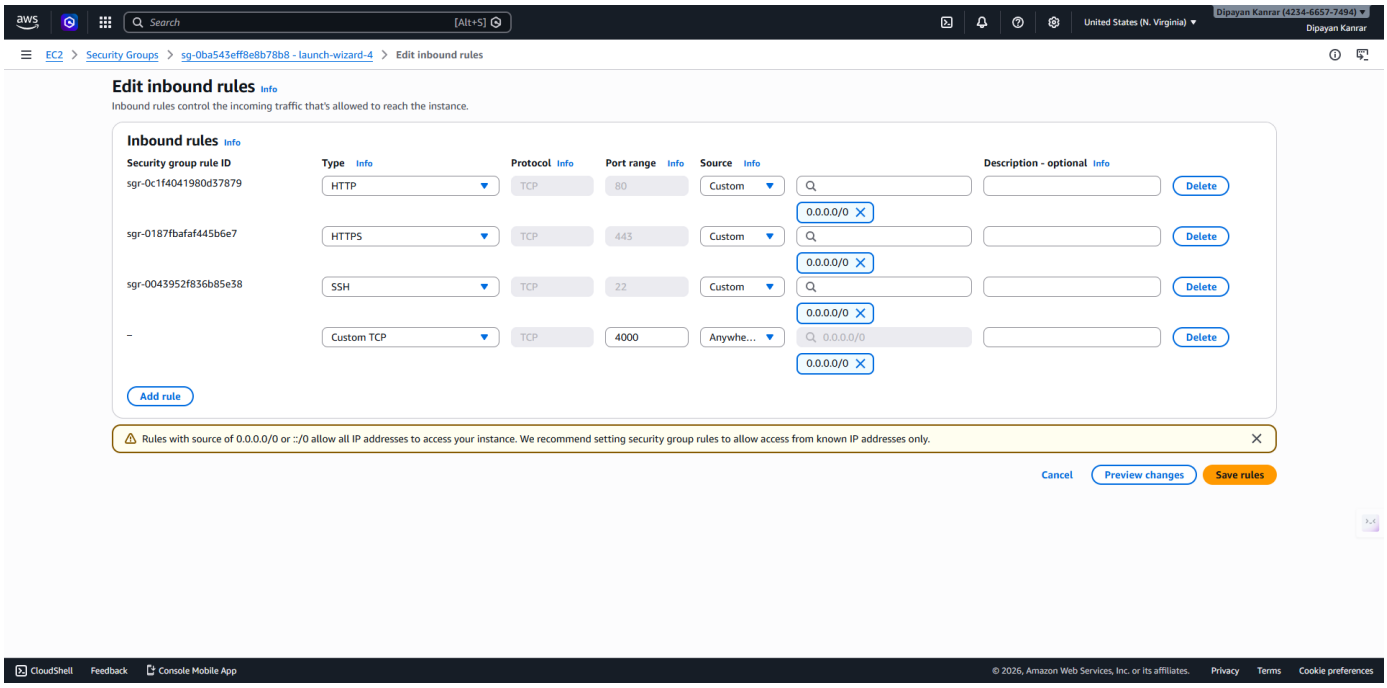
4. Now check whether “Dipayan_ec2”is running or not then click on “Instance ID”.



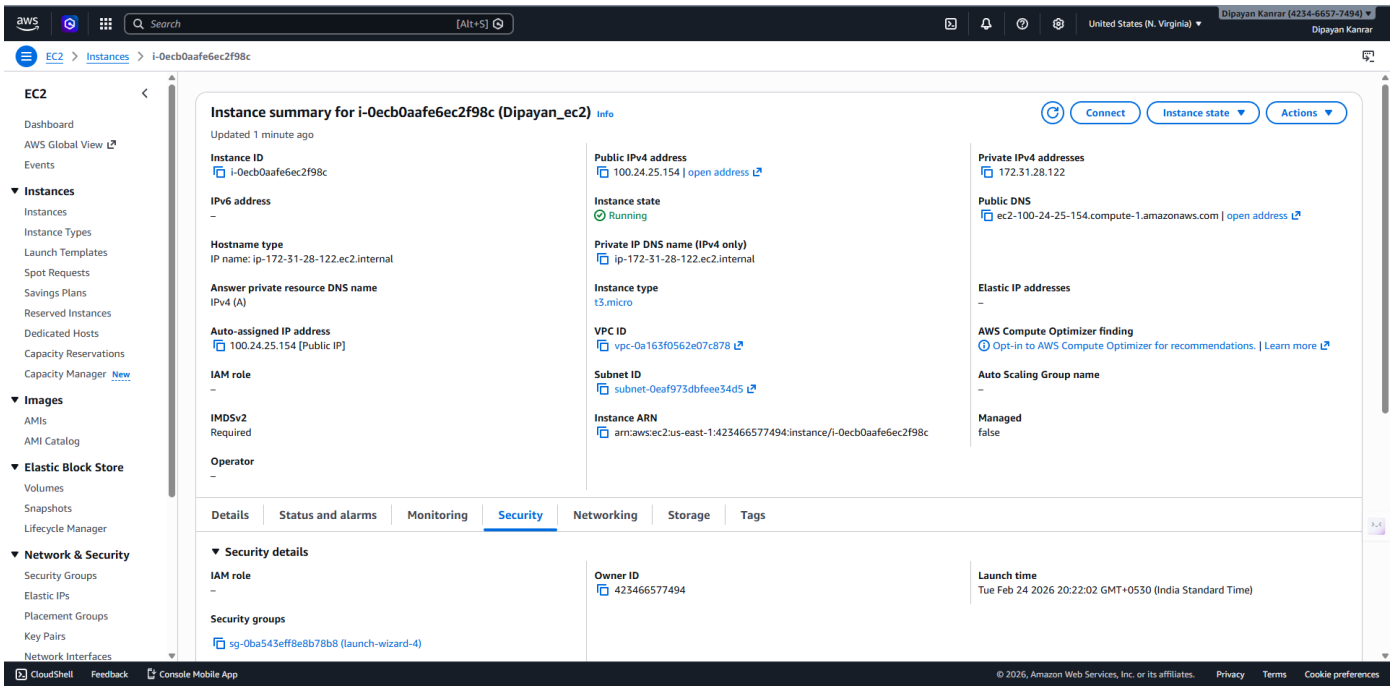
5. Then go to “Security” and click on link under “Security groups” to get the edit option.



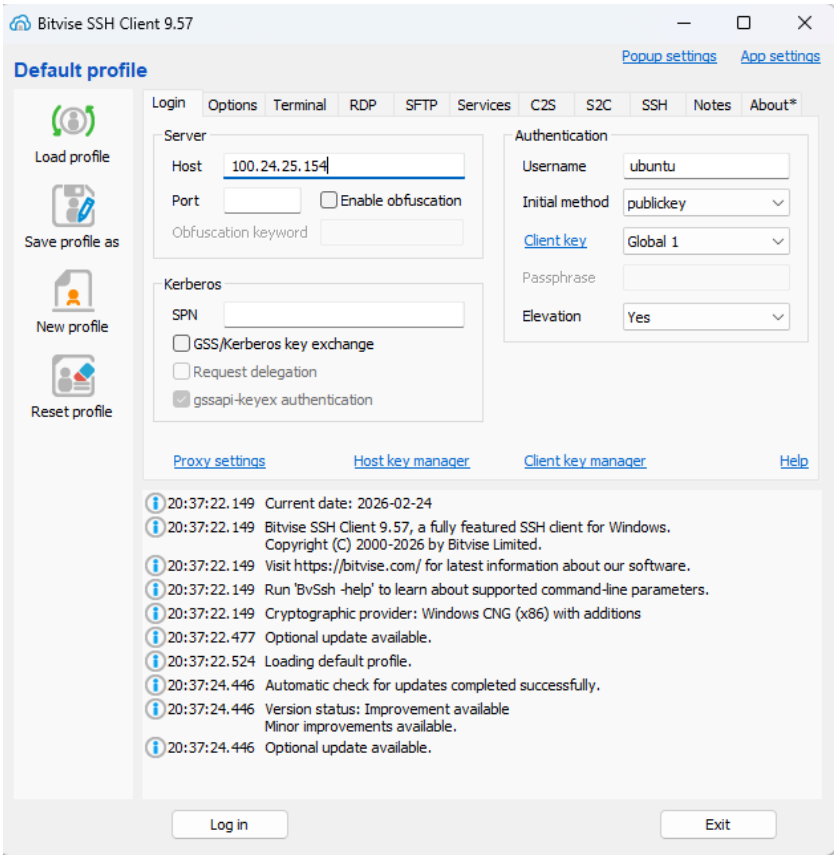
6. Then in “Inbound rules”, click on “Edit inbound rules”, Click on “Add rule”. Set the “Port range” to 4000, in “Source” set “0.0.0.0/0” & click on “Save rules”



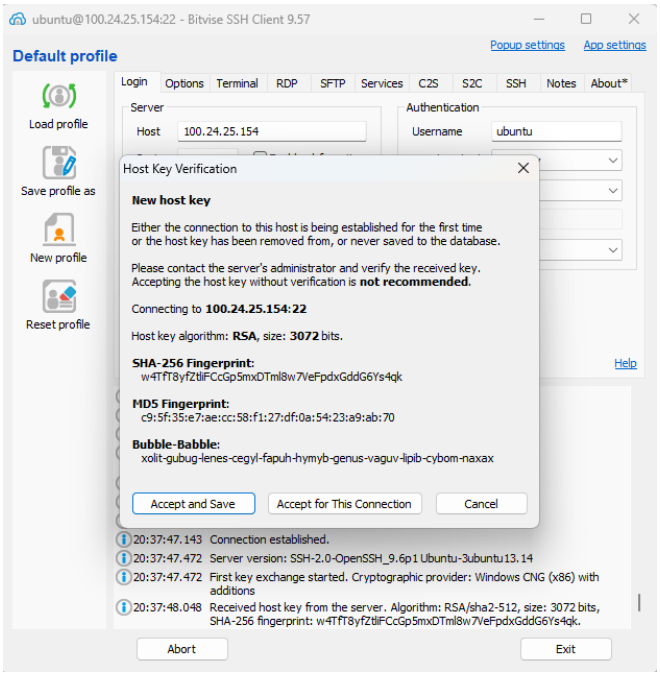
7. Go back to the Instance, copy the “Public IPv4 address”.



8. In “Bitvise SSH Client”, paste the “Public IPv4 address” in “Host” and under “Authentication tab” give the username as Ubuntu, Initial method as “Public key”, Client key as “Global1”. Then click on “Client key manager”.



9. Remove any previously selected key if any, then click on “Import” & select the key with which instance was created. Then click on Log in.



10. After successful “Log in” open a “New Terminal Console”.

11. In the console, type the following commands in sequential order.

```
ubuntu@100.24.25.154:22 - Bitvise xterm - ubuntu@ip-172-31-28-122: ~
/home/ubuntu
ubuntu@ip-172-31-28-122:~$ sudo apt-get update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InReleas
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Pac
Get:5 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe Translati
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Com
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 c-n
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 P
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse Transl
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64
Get:13 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd6
Get:14 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main Tran
Get:15 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd6
Get:16 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd6
Get:17 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe
Get:18 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe
Get:19 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe
Get:20 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe
Get:21 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricte
Get:22 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [
Get:23 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricte
Get:24 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricte
Get:25 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricte
Get:26 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multivers
Get:27 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multivers
Get:28 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multivers
Get:29 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multivers
```

```
ubuntu@ip-172-31-28-122:~$ sudo apt-get upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages have been kept back:
  linux-aws linux-headers-aws linux-image-aws
The following packages will be upgraded:
  base-files bsdextrautils bsdutils cloud-init dirmngr eject fdisk f
  gcc-14-base gir1.2-glib-2.0 gnupg gnupg-l10n gnupg-utils gpg gpg-a
  gpg-wks-client gpgconf gpgsm gpgv inetutils-telnet initramfs-tools
  initramfs-tools-bin initramfs-tools-core keyboxd klibc-utils kpart
  libblkid1 libc-bin libc6 libdrm-common libdrm2 libexpat1 libfdisk1
  libfwupd2 libgcc-s1 libglib2.0-0t64 libglib2.0-bin libglib2.0-data
  libgnutls30t64 libklibc libldap-common libldap2 libmbim-glib4 libm
  libmbim-utils libmount1 libnss-systemd libnuma1 libpam-systemd
  libpng16-16t64 libpython3.12-minimal libpython3.12-stdlib libpytho
  libsmartcols1 libsodium23 libssh-4 libssl3t64 libstdc++6 libsystem
  libsystemd0 libtasn1-6 libudev1 libuuid1 libxml2 libxslt1.1
  linux-tools-common locales motd-news-config mount multipath-tools
  openssl python-apt-common python3-apt python3-distupgrade python3-
  python3-urllib3 python3.12 python3.12-minimal screen snapd systemd
  systemd-dev systemd-hwe-hwdb systemd-resolved systemd-sysv telnet
  ubuntu-release-upgrader-core udev util-linux uuid-runtime
91 upgraded, 0 newly installed, 0 to remove and 3 not upgraded.
Need to get 81.3 MB of archives.
After this operation, 811 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-28-122:~$ sudo apt-get install nginx
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 3 not upgraded.
Need to get 565 kB of archives.
After this operation, 1596 kB of additional disk space will be used.
Do you want to continue? [Y/n]
```

```
ubuntu@ip-172-31-28-122:~$ curl -SL https://deb.nodesource.com/setup_16.x|sudo -E bash -
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100  5736  100  5736    0     0  16619      0 --:--:-- --:--:-- --:--:-- 16626

=============================================================================
=============================================================================

                           DEPRECATION WARNING

    Node.js 16.x is no longer actively supported!

  You will not receive security or critical stability updates for this version.

  You should migrate to a supported version of Node.js as soon as possible.
  Use the installation script that corresponds to the version of Node.js you
  wish to install. e.g.

   * https://deb.nodesource.com/setup_16.x — Node.js 16 "Gallium" (deprecated)
   * https://deb.nodesource.com/setup_18.x — Node.js 18 "Hydrogen" (Maintenance)
   * https://deb.nodesource.com/setup_19.x — Node.js 19 "Nineteen" (deprecated)
   * https://deb.nodesource.com/setup_20.x — Node.js 20 LTS "Iron" (recommended)
   * https://deb.nodesource.com/setup_21.x — Node.js 21 "Iron" (current)



  Please see https://github.com/nodejs/Release for details about which
  version may be appropriate for you.
```

```
ubuntu@ip-172-31-28-122:~$ sudo apt-get install nodejs
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  nodejs
0 upgraded, 1 newly installed, 0 to remove and 3 not upgraded.
Need to get 27.5 MB of archives.
After this operation, 128 MB of additional disk space will be used.
Get:1 https://deb.nodesource.com/node_16.x nodistro/main amd64 nodejs amd64 16.20.2-1nodesource1 [27.
5 MB]
Fetched 27.5 MB in 0s (60.6 MB/s)
Selecting previously unselected package nodejs.
(Reading database ... 71805 files and directories currently installed.)
Preparing to unpack .../nodejs_16.20.2-1nodesource1_amd64.deb ...
Unpacking nodejs (16.20.2-1nodesource1) ...
```

```
ubuntu@ip-172-31-28-122:~$ git clone https://github.com/dev-dipayan809/Project_aws-1.git
Cloning into 'Project_aws-1'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 6 (delta 0), reused 6 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.
ubuntu@ip-172-31-28-122:~$ dir
Project_aws-1
ubuntu@ip-172-31-28-122:~$ cd project_aws-1
-bash: cd: project_aws-1: No such file or directory
ubuntu@ip-172-31-28-122:~$ cd Project_aws-1
ubuntu@ip-172-31-28-122:~/Project_aws-1$
```

```
ubuntu@ip-172-31-28-122:~/Project_aws-1$ npm install

added 224 packages, and audited 225 packages in 13s

26 packages are looking for funding
  run `npm fund` for details

1 low severity vulnerability

Some issues need review, and may require choosing
a different dependency.

Run `npm audit` for details.
npm notice
npm notice New major version of npm available! 8.19.4 -> 11.10.1
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.10.1
npm notice Run npm install -g npm@11.10.1 to update!
npm notice
ubuntu@ip-172-31-28-122:~/Project_aws-1$ node index.js
Started server
```

12. Now copy the “Public IPv4 address” & paste it on a new tab. Add “:4000” at the end and then press enter.

Hello mckv